Compact Modeling of Trap-Assisted Tunneling Current in 3-D NAND Flash Memory

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***Abstract*— In this research, a compact model is proposed for trap-assisted tunneling (TAT) currents in 3-D NAND flash memory during erase/write (EW) cycling. Using the trap spectroscopy by charge injection and sensing (TSCIS) technique, the average trap density and trap energy level are extracted and applied in the TAT model. The compact model integrates band-to-trap tunneling (BT), trap-to-band tunneling (TB), and trap-to-trap tunneling (TTT) mechanisms. In BT and TTT, tunneling to trap occurs only when the trap energy level is aligned with or below the injection level. Modified tunneling equations are used to address misaligned trap energy levels in TTT. The compact model demonstrates good agreement with measurement data across various word-line (WL) voltages and cycling conditions.**

***Index Terms*— 3-D NAND flash memory, bandgap-engineered tunneling oxide (BE-TOX) degradation, compact modeling, endurance, erase/write (EW) cycling, trap-assisted tunneling (TAT), trap spectroscopy by charge injection and sensing (TSCIS).**

I. Introduction

NAND flash memory, a type of nonvolatile memory, is widely used in various electronic devices. Due to challenges related to device scaling and reliability issues, the structure of NAND flash memory has evolved from a 2-D planar to a 3-D vertical architecture [1], [2], [3]. 3-D NAND flash memory consists of a poly-Si channel, a charge trapping layer (CTL), and a bandgap-engineered tunneling oxide (BE-TOX) designed to enhance erase efficiency [4]. A significant challenge in 3-D NAND flash memory is addressing reliability and endurance issues that impact data errors and cell characteristics. Among reliability concerns, trap-assisted tunneling (TAT) is a critical phenomenon in long-term retention [5], [6], [7], [8].

Previous studies on the long-term retention in 3-D NAND flash memory have analyzed and modeled the TAT phenomenon using stretched exponential functions (SEFs) [6], [7] or trap-to-band tunneling (TB) model [8]. However, SEF is an empirical model, and the TB mechanism differs physically from the tunneling that occurs from CTL traps to BE-TOX traps. Therefore, it is necessary to apply a compact model that physically aligns with the phenomenon in TAT.

In this research, TAT current was modeled via band-to-trap tunneling (BT), TB, and trap-to-trap tunneling (TTT) models. The average trap density and trap energy level of the bulk traps in the BE-TOX layer during erase/write (EW) cycling were extracted in our previous study [9] using the trap spectroscopy by charge injection and sensing (TSCIS) technique [10] and were applied to the model. The WL current of the test-element group (TEG) was measured during the EW cycling stress conditions. When the absolute value of the WL voltage ($|V_{\mathrm{WL}}|$) is in a low range, the TAT phenomenon becomes dominant. Therefore, in the measured WL current, the current observed in the low $|V_{\mathrm{WL}}|$ regime was defined and analyzed as TAT current. The tunneling mechanism was analyzed with energy band diagram extracted through technology computer-aided design (TCAD) simulation.

II. Experimental Setup and Results

Fig. 1(a) shows the TEG of a 3-D CTL NAND flash memory, in which approximately 1 000 000 strings are connected in parallel. Each string in the TEG consists of a bitline (BL), a drain select line (DSL), WLs, dummy WLs, a source select line (SSL), and a source line (SL). The WL current is a tunneling current, and for accurate measurements, four WLs are tied to one pad, defined as the selected WL. The endurance test is performed at room temperature using the WL located in the middle of the string. EW cycling is carried out up to 50k cycles, and the WL current is measured after 1, 3, 5, 10, 30, and 50k cycles. Fig. 1(b) shows the waveform used for the EW cycling test. In the erase operation, a gate-induced drain leakage (GIDL) erase mechanism is employed, and a small positive voltage is applied to the unselected and dummy WLs to minimize cycling stress in these WLs. In this test, only a single pulse is applied for both erase and program operations. Both the erase voltage (V_{ERASE}) and program voltage (V_{PGM}) are 20 V.

Fig. 2 presents the WL current measurement results. Fig. 2(a) shows the WL current during a positive sweep of the WL voltage in the erase state, after each EW cycling is completed. Fig. 2(b) shows the WL current when the WL

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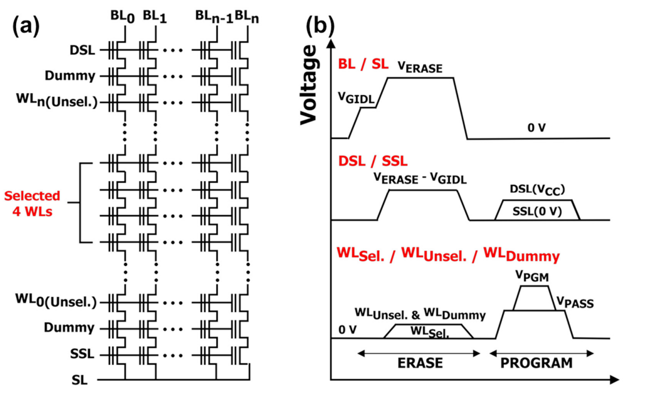

Fig. 1. (a) TEG of 3-D CTL NAND flash memory measured in the experiment. (b) Waveforms for EW cycling stress.

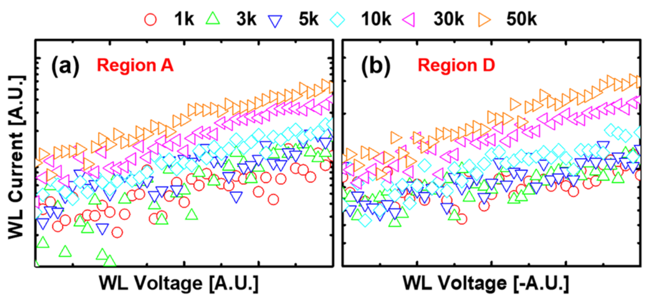

Fig. 2. Measurement results of WL current in the TEG after each EW cycling. (a) WL currents in the low $|V_{WL}|$ range during a positive WL voltage sweep in the erase state. (b) WL currents in the low $|V_{WL}|$ range during a negative WL voltage sweep following the positive sweep.

voltage is swept in the negative direction following the positive sweep. As noted above, because the TAT phenomenon is dominant at low $|V_{WL}|$, only the WL current for the low $|V_{WL}|$ range is shown in Fig. 2. Results at higher $|V_{WL}|$ range are analyzed in detail in [9].

III. Compact Modeling

For modeling and analysis of the TAT mechanism, the conduction band is necessary. In Fig. 3(a), the structure of a single string in the 3-D CTL NAND flash memory is illustrated as applied in the TCAD simulation. The structural parameters of the device are configured to match those of the TEG device used in the measurements. The BE-TOX layer is composed of an oxide/nitride/oxide (O1/N1/O2) stack, and the energy bandgaps of each region are also set to the same values as in the measured device. For the poly-Si channel, the grain boundary (GB) traps are assumed to be located in the middle of each WL and spacer region, following the U-shaped trap distribution shown in Fig. 3(b). A uniform trap distribution is additionally assumed for the channel/BE-TOX and channel/filler interface traps. Consequently, as shown in Fig. 3(c), excellent agreement is observed between the measured BL current and the TCAD simulation results, confirming that the conduction band of this device can be successfully extracted. As shown in Figs. 4 and 5, the conduction band in regions D and A is extracted and analyzed at low $|V_{WL}|$, where the TAT mechanism is dominant.

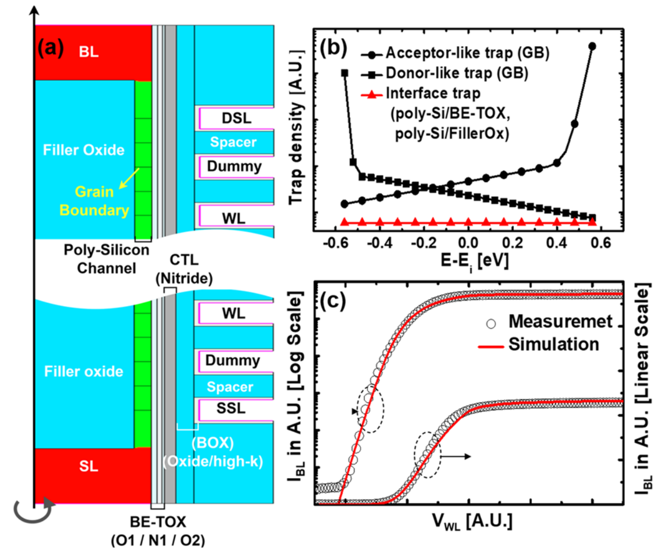

Fig. 3. (a) Cross section of the 3-D CTL NAND flash memory employed in the TCAD simulation. (b) U-shaped trap distribution in the poly-Si channel GB, along with interface traps at the poly-Si/filler oxide and poly-Si/BE-TOX interfaces. (c) Fitting results of the virgin BL current between experimental data and TCAD simulation results.

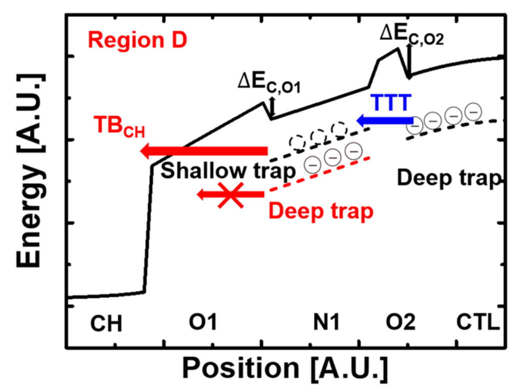

Fig. 4. Conduction band in region D after the positive sweep of the WL voltage, extracted through TCAD simulation.

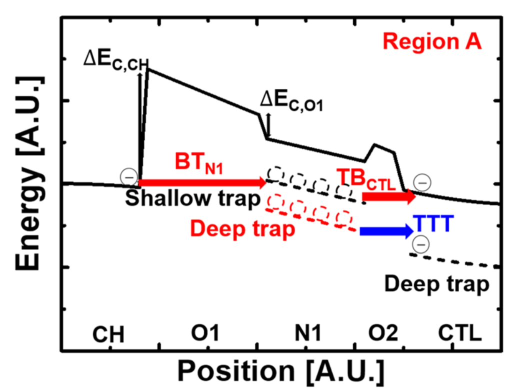

Fig. 5. Conduction band in region A after the erase operation, extracted through TCAD simulation.

Fig. 4 shows the conduction band in region D, when a low $|V_{WL}|$ is applied to the WL after the positive sweep of the WL voltage measurement. After the positive sweep of the WL voltage, electrons become trapped in the CTL, similar to the program state. The number of trapped electrons is determined from the threshold voltage (V_t) and then applied in TCAD simulation. In the CTL, electrons captured in the shallow trap level are emitted into the conduction band via Poole–Frenkel (PF) emission and move laterally [11], making them irrelevant to the WL current. Therefore, only deep trap level is considered in the CTL for the TAT model. When electrons captured in the deep trap level of the CTL, emission to the conduction band via PF is difficult, resulting in TTT from CTL traps to BE-TOX layer traps predominant. Since most of the traps in the BE-TOX layer exist in the N1 layer, it is assumed that the traps exist only in the N1 layer. The TTT model is given as follows [12]:

$$e_{\mathrm{TTT}} = 2\pi^2 \alpha_{\mathrm{N1}} \hbar^3 \left(N_{it,\mathrm{N1}} - n_{it,\mathrm{N1}}\right) \cdot \frac{1}{m_{\mathrm{OX}}^2 t_{\mathrm{O2}} \left(1 + \frac{1}{2\alpha_{\mathrm{CTL}} t_{\mathrm{O2}}}\right)} \exp\left(-\alpha_{\mathrm{CTL,TTT}}\right) \cdot \frac{1}{t_{\mathrm{CTL}}} \int_0^{t_{\mathrm{CTL}}} \exp\left(-\alpha_{\mathrm{CTL},T}\right) dx \frac{1}{t_{\mathrm{N1}}} \int_0^{t_{\mathrm{N1}}} \exp\left(-\alpha_{\mathrm{N1},T}\right) dx \tag{1}$$

where $\hbar$ is the reduced plank constant and m and t are the effective mass and thickness of each layer, respectively. $N_{it,\mathrm{N1}}$ and $n_{it,\mathrm{N1}}$ represent the densities of trap sites and trapped electrons, respectively, which are obtained by converting volume density to interface density. α_{N1} and α_{CTL} are the energy barrier related constants, and $\alpha_{\mathrm{CTL,TTT}}$, $\alpha_{\mathrm{CTL},T}$, and $\alpha_{\mathrm{N1},T}$ are the attenuation coefficients, which associated with the tunneling barriers in each layer, and are given as follows:

$$\alpha_{\mathrm{N1}} = \frac{2\sqrt{2m_N\left(E_{t,\mathrm{CTL}} + \Delta E_{C,\mathrm{O2}} - \Delta E_{C,\mathrm{O1}}\right)}}{\hbar} \tag{2}$$

$$\alpha_{\mathrm{CTL}} = \frac{2\sqrt{2m_N\left(E_{t,\mathrm{CTL}+}\Delta E_{C,\mathrm{O2}}\right)}}{\hbar} \tag{3}$$

$$\alpha_{\mathrm{CTL,TTT}} = \frac{4\sqrt{2m_{\mathrm{OX}}^*}}{3\hbar q F_{O2}} \alpha'_{\mathrm{CTL,TTT}} \times \left[\begin{array}{l} \left(\Delta E_{C,\mathrm{O2}} + E_{t,\mathrm{CTL}} - \beta_{\mathrm{OX}}\sqrt{F_{\mathrm{O2}}}\right)^{3/2} \\ -\left(\Delta E_{C,\mathrm{O2}} + E_{t,\mathrm{CTL}} - \beta_{\mathrm{OX}}\sqrt{F_{\mathrm{O2}}} \right. \\ \left. \qquad -qF_{\mathrm{O2}} t_{\mathrm{O2}}\right)^{3/2} \end{array}\right] \tag{4}$$

$$\alpha_{\mathrm{CTL},T} = \frac{4\sqrt{2m_N^*}}{3\hbar q F_{\mathrm{CTL}}} \left[\left(E_{t,\mathrm{CTL}}\right)^{3/2} - \left(E_{t,\mathrm{CTL}} - qF_{\mathrm{CTL}} x\right)^{3/2}\right] \tag{5}$$

$$\alpha_{\mathrm{N1},T} = \frac{4\sqrt{2m_N^*}}{3\hbar q F_{\mathrm{N1}}} \left[\left(E_{t,\mathrm{N1}}\right)^{3/2} - \left(E_{t,\mathrm{N1}} - qF_{\mathrm{N1}} x\right)^{3/2}\right] \tag{6}$$

$$\beta_{\mathrm{OX}} = \sqrt{q}^3/(\pi\epsilon_{\mathrm{OX}}) \tag{7}$$

where $E_{t,\mathrm{CTL}}$ and $E_{t,\mathrm{N1}}$ are the trap energy level of CTL and N1 layer, respectively, F and ϵ are the average electric field and permittivity of each layer, respectively, and $\alpha'_{\mathrm{CTL,TTT}}$ is a fitting parameter, respectively. In [12], the attenuation coefficients were used in the form of $\alpha \cdot t$, based on the assumption of a rectangular energy barrier without an applied electric field. However, when an electric field is applied, the energy barrier takes on a trapezoidal shape. Therefore, to achieve more accurate physical modeling, the attenuation coefficients $\alpha_{\mathrm{CTL,TTT}}$, $\alpha_{\mathrm{CTL},T}$, and $\alpha_{\mathrm{N1},T}$ are used as shown in (4)–(6), based on the WKB approximation. Additionally, the energy barrier lowering phenomenon due to the electric field induced by the CTL charge is considered by β_{OX} [13]. Electrons trapped in the N1 layer through TTT are emitted to the channel through TB ($\mathrm{TB_{CH}}$). The TB model is shown in Appendix.

Fig. 5 shows the conduction band of region A, when a low positive voltage is applied to the WL after erase operation. In this situation, trapped holes corresponding to V_t are applied in the CTL. During this time, BT occurs from the conduction band of the channel to the traps in the N1 layer. The BT model is given as follows [8], [12]:

$$e_{\mathrm{BT,N1}} = v\mathrm{th}\,\sigma_n\left(N_{t,\mathrm{N1}} - n_{t,\mathrm{N1}}\right) \cdot \exp\left(-\alpha_{\mathrm{O1,BT}}\right) \frac{1}{t_{\mathrm{N1}}} \int_0^{t_{\mathrm{N1}}} \exp\left(-\alpha_{\mathrm{N1},T}\right) dx \tag{8}$$

$$\alpha_{\mathrm{O1,BT}} = \frac{4\sqrt{2m_{\mathrm{OX}}}}{3\hbar q F_{\mathrm{O1}}} \alpha'_{O1,\mathrm{BT}} \times \left[\left(\Delta E_{C,\mathrm{CH}}\right)^{3/2} - \left(\Delta E_{C,\mathrm{CH}} - qF_{\mathrm{O1}} t_{\mathrm{O1}}\right)^{3/2}\right] \tag{9}$$

where $v\mathrm{th}$ is the thermal velocity, σ_n is the capture cross section, and $\alpha'_{\mathrm{O1,BT}}$ is a fitting parameter. Electrons trapped in the N1 layer through BT exhibit different emission mechanisms depending on $E_{t,\mathrm{N1}}$. For the shallow trap level, TB occurs, where electrons move to the conduction band of the CTL ($\mathrm{TB_{CTL}}$). For the deep trap level, TTT occurs, where electrons directly move to traps in the CTL. In the TB and TTT models in region A, the trap densities, energy barrier-related constants, and attenuation coefficients should be adjusted.

IV. Modeling Results and Discussion

A. Trap-Level Dependency

During the EW cycling tests, the anode hole injection (AHI) phenomenon induces degradation of bulk traps in the BE-TOX layer [14], [15]. In our previous study [9], these bulk traps were characterized using the TSCIS method. Fig. 6 shows the average trap density of bulk trap in the BE-TOX layer as a function of trap energy level under various EW cycling conditions. As the number of EW cycles increases, the bulk trap density increases, and additional traps appear at deeper energy levels. Both the density and the energy level of bulk traps significantly influence the TAT phenomenon.

In region D, after the positive sweep of the WL voltage measurement, a significant number of electrons are trapped in the N1 layer and CTL. The $\mathrm{TB_{CH}}$ probabilities of the electrons trapped in the N1 layer vary depending on the trap energy levels. Fig. 7 shows the emission rates of $\mathrm{TB_{CH}}$ and TTT as a function of $E_{t,\mathrm{N1}}$. At the deep trap level of the N1 layer, $\mathrm{TB_{CH}}$ is calculated to be smaller than TTT because electrons are difficult to emit. In this situation, since $N_{it,\mathrm{N1}} - n_{it,\mathrm{N1}} = 0$, TTT does not occur. Therefore, emissions due to TTT are calculated for the shallow trap level of the N1 layer.

Fig. 8 shows the conduction band when $E_{t,\mathrm{N1}}$ is shallow trap. In this case, $E_{t,\mathrm{N1}}$ can be higher than $E_{t,\mathrm{CTL}}$. When $E_{t,\mathrm{N1}}$

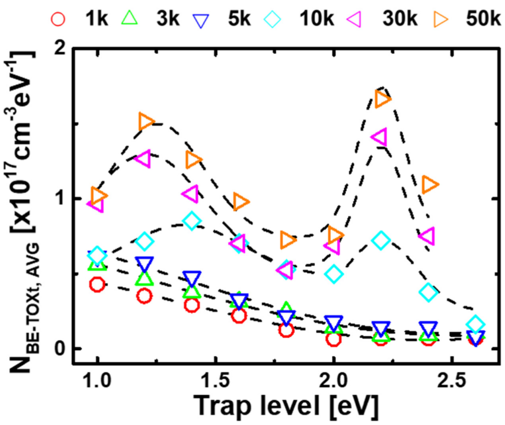

Fig. 6. Average trap density as a function of the conduction band edge in the BE-TOX layer extracted from the trap profile in [9].

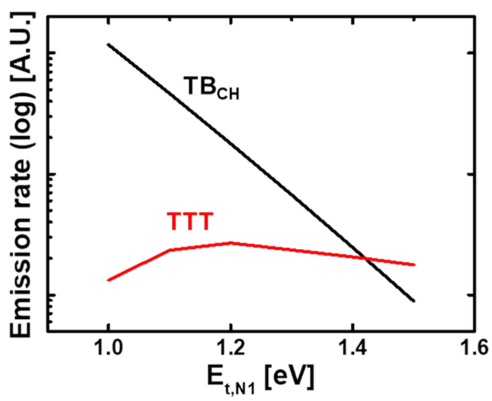

Fig. 7. Emission rate of TB_{CH} and TTT.

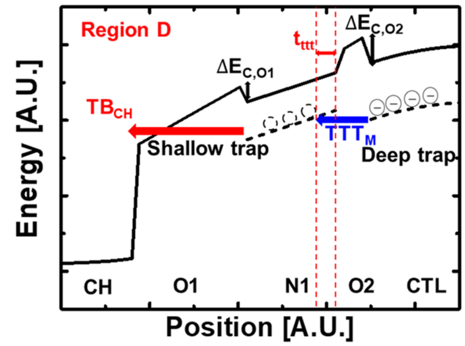

Fig. 8. TTT mechanism with modified barrier and tunneling length when $E_{t,\mathrm{N1}}$ (shallow trap) is higher than $E_{t,\mathrm{CTL}}$ (deep trap).

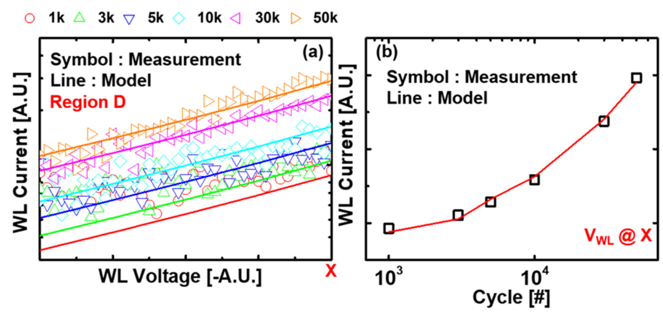

Fig. 9. (a) Compact model results in region D based on WL voltage and EW cycling. (b) Compact model results when the WL voltage is at X.

is higher than $E_{t,\mathrm{CTL}}$, thermal energy is required, significantly reducing the tunneling probability [16]. In addition, according to the inelastic process, tunneling to trap occurs only when the trap energy level is aligned with or below the injection energy level [17], [18], [19], [20]. Therefore, instead of tunneling to the higher trap level, the electrons in the CTL tunnel through a modified barrier and tunneling length to an aligned trap level in the N1 layer. The equations for the modified barrier and tunneling length are as follows [21]:

$$\begin{aligned}
&\alpha_{\mathrm{CTL,TTT},M} \\
&= \frac{4\sqrt{2m_{\mathrm{OX}}}}{3\hbar q F_{\mathrm{O2}}}\alpha'_{\mathrm{CTL,TTT}} \\
&\times \left[\begin{array}{l}
\left(E_{t,\mathrm{CTN}} + \Delta E_{C,\mathrm{O2}} - \beta_{\mathrm{O2}}\sqrt{F_{\mathrm{O2}}}\right)^{3/2} \\
-\left(E_{t,\mathrm{CTN}} + \Delta E_{C,\mathrm{O2}} - \beta_{\mathrm{O2}}\sqrt{F_{\mathrm{O2}}} - qF_{\mathrm{O2}}t_{\mathrm{O2}}\right)^{3/2} \\
+\frac{\epsilon_N}{\epsilon_{\mathrm{OX}}}\sqrt{\frac{m_N}{m_{\mathrm{OX}}}}\left[\begin{array}{c}
\left(E_{t,\mathrm{CTN}} + \Delta E_{C,\mathrm{N1}} - qF_{\mathrm{O2}}t_{\mathrm{O2}}\right)^{3/2} \\
-\left(E_{t,\mathrm{CTN}} + \Delta E_{C,\mathrm{N1}} - qF_{\mathrm{O2}}t_{\mathrm{O2}}\right. \\
\left. - qF_{\mathrm{N1}}t_{\mathrm{TTT}}\right)^{3/2}
\end{array}\right]
\end{array}\right]
\end{aligned} \tag{10}$$

$$t_{\mathrm{N1},M} = t_{\mathrm{N1}} - t_{\mathrm{TTT}} \tag{11}$$

$$t_{\mathrm{O2},M} = t_{\mathrm{O2}} + t_{\mathrm{TTT}}. \tag{12}$$

In region A, after erase operation, a significant number of holes are trapped in the CTL. Additionally, t_{O1} is thicker than t_{O2}, ensuring that $E_{t,\mathrm{N1}}$ in region A is always aligned with or below the conduction band edge of the channel.

When calculating BT_{N1} and TTT, the trap densities for each trap energy level in the N1 layer are determined from the values shown in Fig. 6.

B. Compact Model Results

The WL current is calculated using the following equation:

$$\begin{aligned}
I_{\mathrm{WL,neg}} &= q \cdot \left(e^{-1}_{\mathrm{TB,CH}} + e^{-1}_{\mathrm{TTT}}\right)^{-1} \cdot n_{t,\mathrm{CTL}} \cdot \mathrm{Vol}_{\mathrm{CTL}} \\
&\approx q \cdot e_{\mathrm{TTT,shallow}} \cdot n_{t,\mathrm{CTL}} \cdot \mathrm{Vol}_{\mathrm{CTL}}
\end{aligned} \tag{13}$$

$$\begin{aligned}
I_{\mathrm{WL,pos}} &= q \cdot \left(e^{-1}_{\mathrm{BT,N1}} + e^{-1}_{\mathrm{TB,CTL}} \text{ or } e^{-1}_{\mathrm{TTT}}\right)^{-1} \cdot n_{c,\mathrm{CB}} \cdot \mathrm{Vol}_{\mathrm{CTL}} \\
&\approx q \cdot e_{\mathrm{BT,N1}} \cdot n_{c,\mathrm{CB}} \cdot \mathrm{Vol}_{\mathrm{CTL}}
\end{aligned} \tag{14}$$

where $n_{t,\mathrm{CTL}}$ is calculated from V_t after a positive sweep of the WL voltage, and $n_{c,\mathrm{CB}}$ is the number of electrons in the conduction band of the channel, which is extracted through TCAD simulation. In (14), TTT and TB_{CTL} are calculated to be two or five orders of magnitude larger than BT_{N1}. Therefore, when calculating the TAT current in region A, the tunneling and capture mechanism in the N1 layer through BT_{N1} is dominant.

Fig. 9 shows the fitting results in region D, indicating good agreement between the measurements and the model. It is also confirmed that the model closely matches the measurements at a specific WL voltage denoted as the X point.

Fig. 10 presents the fitting results in region A. When calculating the TAT current in region A, the trap density is evaluated across all trap levels in the BE-TOX N1 layer shown in Fig. 6, resulting in slight discrepancies from the results in region D.

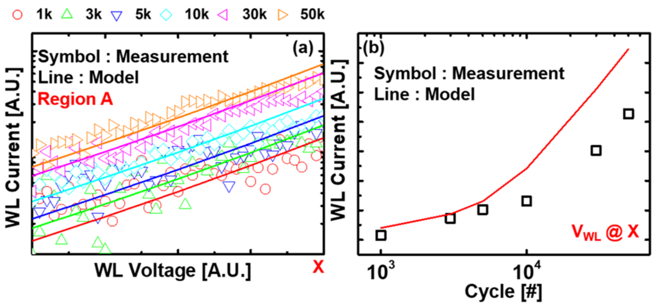

Fig. 10. (a) Compact model results in region A based on WL voltage and EW cycling. (b) Compact model results when the WL voltage is at *X*.

Nevertheless, the overall trends between the measured data and the model agree well.

V. Conclusion

In this research, the TAT currents of 3-D NAND flash memory were modeled using the trap profiles extracted through the TSCIS technique during EW cycling. A compact model was developed integrating the equations for TB, BT, and TTT. In region A, when the trap energy level is aligned with or below the injection level, BT is dominant, resulting in good agreement between the model and measurement data. In region D, modified barrier and tunneling length were applied to account for the dependency of TTT on trap energy levels. Consequently, the model also showed good agreement with the measurement data.

Appendix

The TB model is given as follows [8], [12], [13]:

$$e_{\mathrm{TB}} = \nu_{\mathrm{TB}} \exp\left(-\alpha_{\mathrm{O1,TB}}\right) \frac{1}{t_{\mathrm{N1}}} \int_0^{t_{\mathrm{N1}}} \exp\left(-\alpha_{\mathrm{N1},T}\right) dx \tag{15}$$

$$\alpha_{O1,\mathrm{TB}} = \frac{4\sqrt{2m_{\mathrm{OX}}}}{3\hbar q F_{\mathrm{O1}}} \alpha'_{\mathrm{O1,TB}} \times \left[\begin{array}{l} \left(E_{t,\mathrm{N1}} + \Delta E_{C,\mathrm{O1}} - \beta_{\mathrm{O1}}\sqrt{F_{\mathrm{O1}}}\right)^{3/2} \\ -\left(E_{t,\mathrm{N1}} + \Delta E_{C,\mathrm{O1}} - q F_{\mathrm{O1}} t_{\mathrm{O1}}\right)^{3/2} \end{array} \right] \tag{16}$$

where ν_{TB} is attempt to esca pe frequency and $\alpha'_{\mathrm{TB,CTL}}$ is a fitting parameter.

References

[1] H. Aochi, "BiCS flash as a future 3D non-volatile memory technology for ultra high density storage devices," in *Proc. Int. Memory Workshop*, May 2009, pp. 1–2, doi: 10.1109/IMW.2009.5090581.

[2] H. Kim, S.-J. Ahn, Y. G. Shin, K. Lee, and E. Jung, "Evolution of NAND flash memory: From 2D to 3D as a storage market leader," in *Proc. IEEE Int. Memory Workshop (IMW)*, May 2017, pp. 1–4, doi: 10.1109/IMW.2017.7939081.

[3] K. Parat and A. Goda, "Scaling trends in NAND flash," in *IEDM Tech. Dig.*, Dec. 2018, pp. 2.1.1–2.1.4, doi: 10.1109/IEDM.2018.8614694.

[4] H.-T. Lue et al., "BE-SONOS: A bandgap engineered SONOS with excellent performance and reliability," in *IEDM Tech. Dig.*, Dec. 2005, pp. 547–550, doi: 10.1109/IEDM.2005.1609404.

[5] K. Lee and H. Shin, "Investigation of retention characteristics for trap-assisted tunneling mechanism in sub 20-nm NAND flash memory," *IEEE Trans. Device Mater. Rel.*, vol. 17, no. 4, pp. 758–762, Dec. 2017, doi: 10.1109/TDMR.2017.2772046.

[6] S. Kim, K. Lee, C. Woo, Y. Hwang, and H. Shin, "Analysis of failure mechanisms during the long-term retention operation in 3-D NAND flash memories," *IEEE Trans. Electron Devices*, vol. 67, no. 12, pp. 5472–5478, Dec. 2020, doi: 10.1109/TED.2020.3028349.

[7] Y. Fan, Z. Wang, S. Yang, K. Han, and Y. He, "Investigation of retention characteristics in a triple-level charge trap 3D NAND flash memory," in *Proc. IEEE Int. Rel. Phys. Symp. (IRPS)*, Dallas, TX, USA, Mar. 2022, pp. P31-1–P31-4, doi: 10.1109/IRPS48227.2022.9764506.

[8] J. Park et al., "Decomposition of vertical and lateral charge loss in long-term retention of 3-D NAND flash memory," in *Proc. IEEE Int. Rel. Phys. Symp. (IRPS)*, Monterey, CA, USA, Mar. 2023, pp. 1–4, doi: 10.1109/IRPS48203.2023.10117868.

[9] H. Jo, J. Kim, Y. Cho, H. Shim, J. Sim, and H. Shin, "Investigation of endurance characteristics in 3-D NAND flash memory with trap profile analysis," *IEEE Trans. Electron Devices*, vol. 71, no. 3, pp. 1852–1857, Mar. 2024, doi: 10.1109/TED.2024.3350565.

[10] R. Degraeve et al., "Trap spectroscopy by charge injection and sensing (TSCIS): A quantitative electrical technique for studying defects in dielectric stacks," in *IEDM Tech. Dig.*, Dec. 2008, pp. 1–4, doi: 10.1109/IEDM.2008.4796812.

[11] H.-N. Yoo et al., "Effect of lateral charge diffusion on retention characteristics of 3D NAND flash cells," *IEEE Electron Device Lett.*, vol. 42, no. 8, pp. 1148–1151, Aug. 2021, doi: 10.1109/LED.2021.3088851.

[12] Y. Wang and M. H. White, "An analytical retention model for SONOS nonvolatile memory devices in the excess electron state," *Solid-State Electron.*, vol. 49, no. 1, pp. 97–107, Jan. 2005.

[13] M. Kim and H. Shin, "Analysis and compact modeling of fast detrapping from bandgap-engineered tunneling oxide in 3-D NAND flash memories," *IEEE Trans. Electron Devices*, vol. 68, no. 7, pp. 3339–3345, Jul. 2021.

[14] T. Samadder, N. Choudhury, S. Kumar, D. Kochar, N. Parihar, and S. Mahapatra, "A physical model for bulk gate insulator trap generation during bias-temperature stress in differently processed p-channel FETs," *IEEE Trans. Electron Devices*, vol. 68, no. 2, pp. 485–490, Feb. 2021, doi: 10.1109/TED.2020.3045960.

[15] S. Kumar, R. Anandkrishnan, N. Parihar, and S. Mahapatra, "A stochastic framework for the time kinetics of interface and bulk oxide traps for BTI, SILC, and TDDB in MOSFETs," *IEEE Trans. Electron Devices*, vol. 67, no. 11, pp. 4741–4748, Nov. 2020, doi: 10.1109/TED.2020.3020533.

[16] K. Lee et al., "Activation energies (E_a) of failure mechanisms in advanced NAND flash cells for different generations and cycling," *IEEE Trans. Electron Devices*, vol. 60, no. 3, pp. 1099–1107, Mar. 2013, doi: 10.1109/TED.2013.2241065.

[17] S. Takagi, N. Yasuda, and A. Toriumi, "A new I–V model for stress-induced leakage current including inelastic tunneling," *IEEE Trans. Electron Devices*, vol. 46, no. 2, pp. 348–354, Feb. 1999.

[18] D. Ielmini, A. S. Spinelli, M. A. Rigamonti, and A. L. Lacaita, "Modeling of SILC based on electron and hole tunneling. I. Transient effects," *IEEE Trans. Electron Devices*, vol. 47, no. 6, pp. 1258–1265, Jun. 2000, doi: 10.1109/16.842971.

[19] E. Vianello et al., "Experimental and simulation analysis of program/retention transients in silicon nitride-based NVM cells," *IEEE Trans. Electron Devices*, vol. 56, no. 9, pp. 1980–1990, Sep. 2009, doi: 10.1109/TED.2009.2026113.

[20] Z.-Y. Lun, Y. Li, K. Zhao, G. Du, X.-Y. Liu, and Y. Wang, "Modeling of trap-assisted tunneling on performance of charge trapping memory with consideration of trap position and energy level," *Chin. Phys. B*, vol. 25, no. 8, Aug. 2016, Art. no. 088502, doi: 10.1088/1674-1056/25/8/088502.

[21] H.-T. Lue et al., "Understanding barrier engineered charge-trapping NAND flash devices with and without high-K dielectric," in *Proc. IEEE Int. Rel. Phys. Symp.*, Apr. 2009, pp. 874–882, doi: 10.1109/IRPS.2009.5173370.